

Final Project Report: AI-Driven Reservoir Performance Proxy

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1- Problem Statement:

The oil and gas industry relies heavily on numerical reservoir simulators (like OPM Flow) to predict long-term production behaviour and assess the economic viability of drilling projects. However, these traditional physics-based simulators solve complex differential equations over millions of grid cells, making them computationally expensive. A single 22-year scenario can take minutes to hours to evaluate, which fundamentally bottlenecks real-time decision-making, history matching, and Net Present Value (NPV) optimization.

2- Objective:

This project aims to build an AI-driven proxy model capable of learning the underlying thermodynamics and fluid mechanics from simulation data. By acting as a highly accurate surrogate, the proxy will instantly (in milliseconds) predict 22-year oil production rates and reservoir pressures given varying geological and operational parameters.

Model	Type	Primary Objective
Random Forest	Ensemble (baseline)	Scalar prediction + feature importance
MLP	Deep neural network	Non-linear scalar prediction
PINN	Physics-informed NN	Physically valid scalar prediction
Enc-Dec LSTM	Sequence model	Full 22-year timeseries prediction

3- Input Parameters:

Six parameters were varied across 200 simulations using Latin Hypercube Sampling (LHS) to maximise parameter space coverage:

- Producer Bottom Hole Pressure (BHP) — controls drawdown and flow rate
- Gas Injection Rate — volume of gas injected to maintain reservoir pressure
- Injector BHP Limit — maximum allowable injection pressure (safety ceiling)
- Initial Production Period — days before gas injection begins
- Permeability Multiplier — geological property controlling fluid flow through rock
- Porosity Multiplier — geological property controlling storage capacity

4- Output Targets:

The scalar models (RF, MLP, PINN) predict four end-state values after 22 years. The LSTM additionally predicts the full trajectory of eight physical variables at every timestep:

- final_fopt — Cumulative oil production (STB) — primary revenue KPI
- final_fpr — Final reservoir pressure (PSIA) — depletion planning KPI
- final_fopr — Final oil rate (STB/day) — well longevity KPI

- peak_fopr — Peak oil rate (STB/day) — facility sizing KPI
- LSTM additionally: fgir, fgit, fgpr, fgpt, wbhp_inj at every timestep

5- Assumptions and Hypotheses about Data and Model:

5.1 Data Assumptions:

- The 200 SPE10 benchmark simulations are representative of the parameter space. Latin Hypercube Sampling ensures uniform coverage across all six input dimensions.
- Each simulation run is statistically independent. There is no temporal dependency between different simulation runs (only within a single run for the LSTM).
- The OPM Flow simulator output is deterministic and noise-free. No data cleaning for measurement noise was required.
- The six input parameters capture the dominant sources of variability. External factors such as fluid viscosity and rock compressibility are assumed constant at SPE10 benchmark values.
- Negative production or pressure values are physically impossible under normal operating conditions and would be removed as outliers before any real-field deployment.

5.2 Model Assumptions:

- The relationship between the 6 input parameters and the 4 scalar outputs is smooth and learnable from 200 examples, validated by RF achieving $R^2=0.988$ on cumulative oil.
- For the LSTM: the encoder context vector (latent_dim=256) is sufficient to represent the full geological and operational fingerprint of a scenario.
- For the PINN: reservoir physics constraints can be encoded as loss function penalties without requiring full PDE discretisation. Specifically, Darcy's Law and injection pressure limits provide meaningful engineering guardrails.
- StandardScaler (zero mean, unit variance) is the appropriate normalisation for output targets. MinMaxScaler is used for bounded inputs (6 parameters and time axis).
- The adaptive time-stepping of OPM Flow (irregular, physics-driven timestep spacing) carries meaningful information and should NOT be interpolated to a uniform grid.

5.3 Hypotheses:

- H1 (Feature Importance): Geological parameters (permeability, porosity) will dominate over operational parameters in driving production variability confirmed: geology accounts for 87%.
- H2 (Architecture): Simpler neural architectures will outperform deeper ones on this small dataset (200 rows) confirmed: MLP 64→32 outperforms 128→64→32.
- H3 (Physics Constraints): Adding reservoir engineering constraints to the loss function will improve prediction reliability without sacrificing accuracy partially confirmed: PINN achieves 95% constraint compliance at comparable accuracy to MLP.

- H4 (Timeseries): The Encoder-Decoder LSTM will successfully generalise across 200 independent scenarios with variable-length sequences confirmed: fopr $R^2=0.984$ on 30 held-out test runs.

6- Exploratory Data Analysis (EDA):

6.1 Dataset Overview:

Two datasets were compiled from the 200 OPM Flow simulations:

Dataset	Rows	Columns	Purpose
ml_dataset_scalar.csv	200	17	RF, MLP, PINN training (scalar targets)
ml_dataset_timeseries_for_lstm.csv	158,742	18	LSTM training (full timeseries)

6.2 Key EDA Findings:

Feature Distributions

All six input parameters follow uniform distributions by construction (Latin Hypercube Sampling). Histograms confirmed no clustering or gaps in parameter coverage, validating the sampling strategy.

Correlation Analysis:

A Pearson correlation heatmap on the scalar dataset revealed:

- Permeability multiplier shows the strongest positive correlation with cumulative oil (fopt) and final pressure (fpr).
- Porosity multiplier shows the second strongest positive correlation.
- Operational parameters (BHP, injection rate) show weaker but meaningful correlations.
- Combined, geological parameters explain ~87% of output variability (confirmed by RF feature importance).

Timeseries Structure:

Analysis of the timeseries dataset revealed critical structural characteristics:

- Each of the 200 simulation runs has a different number of timesteps (range: 742–848 steps).
- Timestep spacing is irregular and adaptive OPM Flow takes small steps (as small as 0.67 days) during rapid pressure transients early in production, and larger steps (up to 30 days) during stable late-stage production.
- This irregular spacing is physically meaningful and carries information about reservoir dynamics. It was NOT interpolated to a uniform grid to preserve this signal.

Output Variable Ranges:

Output variables span multiple orders of magnitude, requiring careful normalisation:

- fgit (cumulative gas injected): 0.65 – 358,107 MSCF
- fopr (oil rate): 0 – 1,202 STB/day
- fpr (reservoir pressure): 2,512 – 10,000 PSIA

This 1,000× scale difference motivated per-variable StandardScaler normalisation for neural network targets, preventing large-range variables from dominating the loss function.

FPR Oscillation Pattern:

Visual inspection of FPR (reservoir pressure) timeseries revealed rapid oscillations after approximately day 3,000 in many simulations runs. These oscillations reflect real cyclic gas injection physics: when injection is on, pressure rises; when injection is off, pressure declines. This high-frequency signal is real OPM Flow output, not measurement noise. The LSTM was expected (and confirmed) to learn the smooth pressure depletion envelope rather than fit the individual injection cycles.

7- Feature Engineering & Data Transformations:

Given the significant scale differences between geological multipliers and fluid volumes, we normalized all variables using Standard Scaler to stabilize gradient descent.

7.1- Handling the Sequence Complexity:

The numerical simulator uses an adaptive time stepping algorithm (taking tiny time steps during rapid pressure changes and large steps during steady state). To feed this into an LSTM, we engineered an innovative sequence handling pipeline:

- **Sentinel Padding:** Shorter simulation sequences were padded to the maximum length using a - 1.0 sentinel value (safely outside the scaled range of real data).
- **Masking Layers:** A Keras Masking layer and custom sample_weight matrices were implemented so the neural network completely ignored the padded time steps during both state propagation and loss calculation.
- **Historical Grid Inference:** We generated an avg_time_grid from the training distribution to ensure that new inference requests exactly matched the simulator's learned adaptive spacing.

8- Proposed Approaches & Overfitting/Underfitting Analysis:

8.1- Data Split Strategy:

All models used a run-level split (not row-level) to prevent data leakage in the timeseries dataset:

- RF: 80/20 train/test split (160/40 runs)
- MLP, PINN, LSTM: 70/15/15 train/val/test split (140/30/30 runs)

Cross-validation for the RF was performed strictly on the 160 training runs, not the full 200-run dataset.

8.2- Random Forest (Baseline):

Hyperparameters:

- `n_estimators=500`, `max_features=None`, `min_samples_leaf=2`
- Key finding: `max_features=None` (use all 6 features) outperforms `max_features='sqrt'`. With only 6 features, restricting each tree to $\sqrt{6} \approx 2-3$ features is overly aggressive.

Overfitting Analysis:

The train vs test R^2 gap (a direct measure of overfitting) was +0.004 for cumulative oil effectively zero. Random Forest is naturally protected against overfitting through the bagging effect: averaging 500 trees trained on different bootstrap samples cancels individual tree memorisation. The gap of +0.004 is well below the concern threshold of 0.05.

8.3 - MLP (Deep Neural Network Baseline):

Architecture Selection:

Architecture search empirically demonstrated that simpler networks generalise better on this 200-row dataset:

Architecture	Parameters	Test R^2 (fopt)	Test R^2 (fpr)	Gap
128→64→32→4 (original)	11,682	0.990	0.896	+0.033
64→32→4 (final)	~2,660	0.996	0.972	+0.006

The simpler 64→32 architecture achieves higher accuracy AND a smaller overfitting gap. This counter-intuitive result is physically motivated: with 140 training samples, 11,682 parameters create ~83 parameters per training sample, which encourages memorisation. The 64→32 architecture at ~19 parameters per sample forces the model to learn general patterns.

Regularisation

- Dropout(0.1) applied after each hidden layer
- BatchNormalization was empirically removed — it degraded performance by over-normalising the 6 input features in this small-data regime
- EarlyStopping(patience=50) with `restore_best_weights=True`
- ReduceLROnPlateau(factor=0.5, patience=20, min_lr=1e-6)
- Huber loss (more robust than MSE to physical outliers in extreme geological scenarios)

8.4 - PINN (Physics-Informed Neural Network):

Physics Constraint Design

The PINN uses the same 64→32 base architecture as the final MLP, plus a physics penalty term in the loss function:

$$\text{Total Loss} = \text{Huber Data Loss} + \lambda \times \text{Physics Penalty}$$

Three constraints were designed and empirically validated:

- Constraint 1 (Material Balance): Initially included, then removed after data analysis showed cross-sectional FOPT-FPR correlation across 140 independent scenarios = -0.071 (near zero). Batch correlation on 16-sample mini-batches swung from -0.959 to $+0.984$ pure statistical noise, not physics.
- Constraint 2 (Darcy's Law): FPR must exceed producer BHP. If reservoir pressure falls below wellbore pressure, flow reversal occurs physically impossible. Validated: 9/140 training scenarios approach this boundary, confirming it is a meaningful constraint. Computed in real PSI units (not scaled space) to avoid cross-scaler mismatch.
- Constraint 3 (Injection Ceiling): FPR cannot exceed injector BHP limit. Gas injection stops when reservoir pressure reaches the injector's operating ceiling. Validated: 0/140 training violations a hard physical ceiling always respected. Computed in real PSI units.

Lambda Sensitivity:

Lambda (λ) controls the strength of physics enforcement relative to data fit:

- $\lambda=0.001$: fo_{pt} $R^2=0.998$, fpr $R^2=0.934$ — physics barely active
- $\lambda=0.01$ (selected): fo_{pt} $R^2=0.996$, fpr $R^2=0.960$ — balanced
- $\lambda=0.1$: fo_{pt} $R^2=0.994$, fpr $R^2=0.920$ — stronger physics, slight accuracy trade-off
- $\lambda=1.0$: fo_{pt} $R^2=0.995$, fpr $R^2=0.924$ — heavy physics, still manageable

$\lambda=0.01$ was selected as the optimal balance. Importantly, after fixing the constraint design (removing Constraint 3, fixing unit mismatch), the lambda sensitivity became well-behaved — earlier versions with the flawed constraints showed catastrophic collapse (fpr $R^2=0.35$) at $\lambda=0.1$.

8.5 - Encoder-Decoder LSTM:

Architecture:

- Encoder: Dense(64, linear) → BatchNorm → ReLU → Dense(128, linear) → BatchNorm → ReLU → h,c context states
- Context: latent_dim=256, producing separate h and c states for each LSTM decoder layer
- Masking(mask_value=-1.0): skips padded timesteps before first LSTM layer
- Decoder LSTM 1: 256 units, return_sequences=True, Dropout(0.2)
- Decoder LSTM 2: 128 units, return_sequences=True, Dropout(0.1)

- Output: TimeDistributed(Dense(8, linear)), predicts all 8 variables at every timestep

Loss Function:

A custom weighted Huber loss was implemented to prevent large-range variables from dominating gradients:

LOSS_WEIGHTS = [1.0, 1.0, 1.0, 1.0, 1.0, 1.5, 1.5, 1.0] for [fgir, fgit, fgpr, fgpt, fopr, fopt, fpr, wbbp_inj] fopt and fpr received 1.5× weight because these cumulative/integral variables are inherently harder to predict than rates.

Training Protocol:

- Adam(lr=1e-3, clipnorm=1.0) — gradient clipping prevents the gradient explosion observed in early runs (epoch 58 spike)
- EarlyStopping(patience=40, restore_best_weights=True)
- ReduceLROnPlateau(factor=0.5, patience=10, min_lr=1e-6)
- batch_size=8 — smaller batches provide more gradient updates per epoch
- Inference: average historical adaptive time grid (derived from training runs) used for new scenario prediction avoids out-of-distribution time inputs

9- Proposed Solution — Model Selection:

9.1 - Selection Criteria:

Models were evaluated on four criteria:

- Accuracy: R^2 on held-out test set (30 runs for neural models, 40 for RF)
- Consistency: CV R^2 vs test R^2 gap (measure of overfitting)
- Physical validity: compliance with reservoir engineering constraints
- Unique capability: what each model provides that others cannot

9.2 - Scalar Model Comparison (RF vs MLP vs PINN):

Target	RF R^2	MLP R^2	PINN R^2	Business Application
fopt (Cumul. Oil)	0.988	0.996	0.996	22-year revenue forecast
fpr (Pressure)	0.934	0.972	0.960	Depletion & CAPEX planning
fopr (Final Rate)	0.972	0.964	0.942	Well longevity assessment
peak_fopr	0.982	0.984	0.981	Facility & OPEX sizing

PINN wins on none of the pure accuracy metrics but uniquely provides 95% physics constraint compliance.

9.3 - Selected Model: PINN (Primary) + LSTM (Extended):

Why PINN over MLP:

Both MLP and PINN achieve $R^2=0.996$ on cumulative oil the most important business KPI. The PINN is selected as the primary scalar model because:

- Physical reliability: 95% compliance with Darcy's Law and injection BHP limits across all test runs. Zero impossible pressure predictions.
- Engineering defensibility: predictions can be verified against physical laws, making the model auditable for real deployment.
- Intellectual contribution: the empirical validation methodology (removing the flawed material balance coupling, fixing unit mismatches) demonstrates rigorous physics-informed design beyond a textbook PINN implementation.

Why LSTM (Separate Category):

The LSTM is not directly comparable to the scalar models because it solves a fundamentally different and harder problem: predicting the full 22-year trajectory rather than end-state scalars. It is selected as the timeseries extension because:

- fopr $R^2=0.984$ — the highest oil rate accuracy of any model in the project
- 6 out of 8 physical variables achieve $R^2 > 0.94$
- Uniquely enables the future interactive dashboard given any 6 input parameters, instantly generate the full 22-year production curve
- Adaptive inference grid maintains the model in its natural distribution for new scenario prediction

10- Results & Learnings:

10.1 - Final Model Results:

Scalar Models:

Model	fopt R^2	fpr R^2	fopr R^2	peak_fopr R^2
Random Forest	0.988	0.934	0.972	0.982
MLP (final)	0.996	0.972	0.964	0.984
PINN (final)	0.996	0.960	0.942	0.981

LSTM Timeseries Model:

Variable	R ²	MAE	Unit	Tier
fopr (oil rate)	0.984	3.98	STB/D	Excellent
fgit (cumul. gas inj)	0.991	4,656	MSCF	Excellent
fgpt (cumul. gas prod)	0.990	4,665	MSCF	Excellent
fgpr (gas rate)	0.951	1.09	MSCF/D	Excellent
fgir (gas inj rate)	0.947	1.09	MSCF/D	Excellent
fopt (cumul. oil)	0.965	1,976	STB	Good
wbhp_inj (injector BHP)	0.920	302	PSIA	Good
fpr (pressure)	0.842	249	PSIA	Weakest*

*FPR R²=0.842 reflects the cyclic gas injection oscillations present in OPM Flow output. The LSTM correctly learns the smooth pressure depletion envelope without overfitting to high-frequency injection cycle noise the physically relevant signal for engineering decisions.

10.2 - Key Learnings:**Learning 1: Geology dominates operations (87% vs 13%):**

The RF feature importance analysis revealed that permeability multiplier (0.474) and porosity multiplier (0.400) together account for 87% of cumulative oil production variability. All six operational parameters combined contribute only 13%. Business implication: investment in geological surveys and core sampling before field development is far more valuable than operational optimisation for this reservoir type.

Learning 2: Simpler architectures generalise better on small datasets:

The empirical architecture comparison (128→64→32 vs 64→32) demonstrated that for 200-row datasets, smaller models achieve higher test accuracy AND smaller overfitting gaps. The train-test R² gap shrank from +0.033 to +0.006 by reducing the architecture. This counter-intuitive result is consistent with statistical learning theory: with 140 training samples, a 11,682-parameter model has ~83 parameters per sample, which incentivises memorisation.

Learning 3: Physics constraints must be empirically validated:

The material balance coupling constraint was initially included because textbook reservoir physics states that ‘high production + high pressure is suspicious.’ However, data analysis revealed:

- Cross-sectional FOPT-FPR correlation across 140 independent scenarios = -0.071 (near zero, not physically meaningful)
- Mini-batch correlation on 16 samples swings from -0.959 to $+0.984$ (pure statistical noise)
- High-permeability scenarios legitimately sustain both high production AND high pressure through efficient gas injection

Removing Constraint 3 improved FPR R^2 from 0.350 to 0.960 in the PINN. This demonstrates that physics constraints must be validated against the specific reservoir geometry and dataset, not applied from textbook assumptions.

Learning 4: The LSTM solves a qualitatively different problem:

The LSTM is not a more accurate version of the scalar models. It answers a different question. Scalar models ask ‘what is the state after 22 years?’ The LSTM asks, ‘what is the complete physical trajectory?’ This distinction has direct engineering value: knowing that pressure peaks in year 3 rather than year 10 changes the entire field development plan, even if the final pressure value is the same.

Learning 5: Gradient engineering is critical for physics-informed training:

Three specific gradient-related issues were identified and resolved during PINN development:

- Gradient explosion: Early LSTM training showed a spike at epoch 58 where val_loss jumped from 0.18 to 0.93. Fixed by clipnorm=1.0 in the Adam optimizer.
- Gradient scale mismatch: Physics penalties using MSE (quadratic) against Huber data loss (linear for large errors) caused $6\times$ gradient imbalance at large violations. Fixed by using Huber for both.
- Cross-scaler comparison: Physics constraints comparing fpr_pred (scaled by scaler_y) against BHP inputs (scaled by scaler_X) in the same normalised space caused 91/140 false violations. Fixed by unscaling to real PSI units before comparison.

11- Future Work:

While this proxy successfully predicts fluid dynamics across operational and geological boundaries, it can be extended:

- **Time-Series PINN:** Apply the scalar PINN constraints directly across the internal time steps of the LSTM to strictly enforce monotonicity across the entire sequence.
- **Economic Integration:** Connect the proxy output (Production curves) directly into a financial Net Present Value (NPV) calculation pipeline, creating an end-to-end investment optimization tool.